

# StudyMate: A Developer Collaboration Platform

Group: Web Crew

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# Brief about the Project

**Problem:** Students and developers often struggle to find dedicated coding partners with similar skill sets and goals for projects.

**Solution:** *StudyMate* is a streamlined social platform designed exclusively for developers.

## Core Features:

- **Find Partner:** An algorithm to discover and connect with like-minded developers based on tech stacks.
- **Community Posts:** A dedicated feed to share project updates, ask technical questions, and showcase work.
- **Developer Profiles:** Customizable profiles highlighting individual skills and current focuses.

# Comparison with Existing Systems

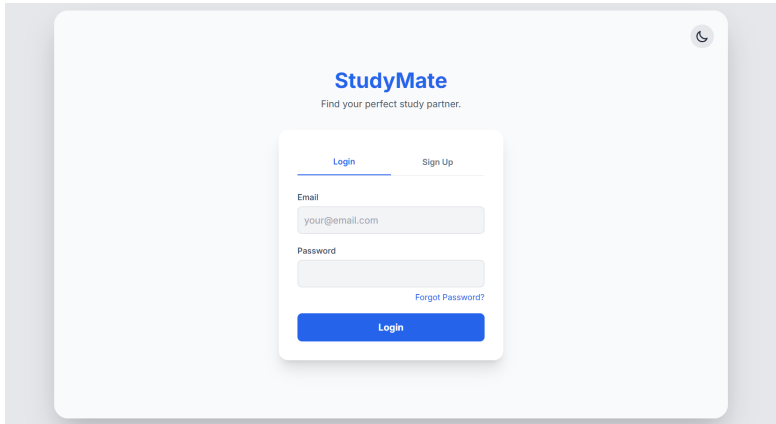
| Feature       | General Social Media           | StudyMate (Our Project)            |
|---------------|--------------------------------|------------------------------------|
| Primary Focus | Broad networking, job hunting. | Focus on finding project partners. |
| Matchmaking   | Manual searching.              | Tech-stack specific matching.      |
| Distractions  | High (ads, viral posts).       | Low (strictly developer-centric).  |

# Tools Being Used

This project is being developed using the **MERN Stack** for a robust, scalable, and responsive application.

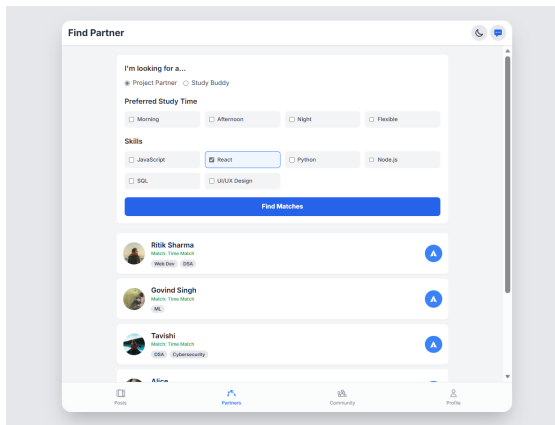
- **Frontend:** React.js (built with Vite), Tailwind CSS for rapid UI styling.
- **Backend:** Node.js with Express.js for handling API routes and business logic.
- **Database:** MongoDB (NoSQL) for flexible data storage.
- **Authentication:** JSON Web Tokens (JWT) for secure user sessions.

# User Interface: Secure Authentication



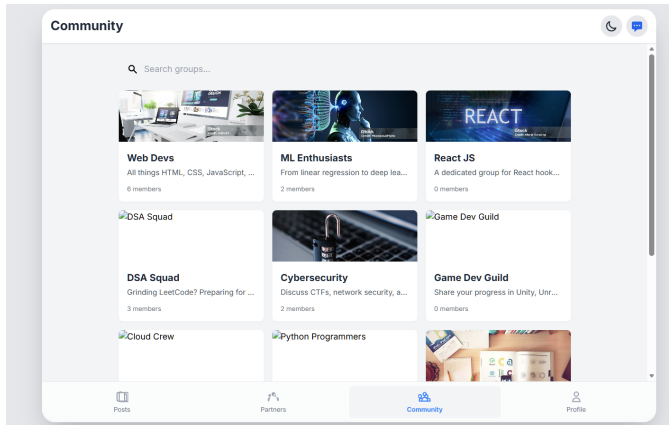
- JWT-based secure login and signup system.
- Responsive design for seamless access across devices.

# User Interface: Smart Partner Discovery



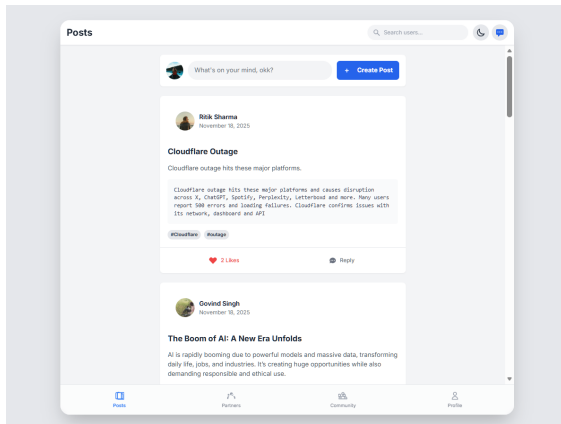
- Filter partners by **Stack** (React, Node, etc.) and **Time Preference**.
- Direct connection matching for project collaboration.

# User Interface: Interest-Based Groups



- Explore niche developer communities like **ML Enthusiasts** and **DSA Squad**.
- Join groups to discuss specific tech-stack challenges.

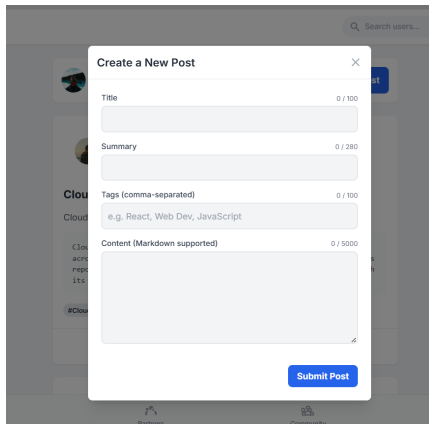
# User Interface: Real-time Discussion Feed



- Active feed for sharing technical news and project updates.
- Interactive "Like" and "Reply" features for developer engagement.



# User Interface: Markdown Supported Posting



The image shows a 'Create a New Post' modal form overlaid on a blurred background of a social media feed. The form has a title bar with a close button (X). It contains four input fields: 'Title' (0 / 100), 'Summary' (0 / 280), 'Tags (comma-separated)' (0 / 100) with a placeholder 'e.g. React, Web Dev, JavaScript', and 'Content (Markdown supported)' (0 / 5000). A blue 'Submit Post' button is at the bottom right of the form. The background shows a search bar, user avatars, and a list of posts with titles like 'Cloud', 'Cloud', and 'Cloud'.

- Advanced post editor with **Markdown support** for code snippets.
- Tagging system (`#Cloud`, `#Outage`) for better discoverability.

# Future Scope & Planned Features

- **Real-Time Chat:** Implement Socket.io to allow direct, real-time messaging between matched partners.
- **GitHub Integration:** Automatically fetch and display users' top repositories and recent commit activity on their profiles.
- **AI Matchmaking:** Upgrade the matching algorithm using machine learning to suggest partners based on coding habits and past project success.

# Thank You!